

Start together

Prepare the materials

Use the small (1×) pieces from the 21st Century and Upscale Pattern Blocks. Put larger pieces aside. Print US Letter, single-sided, at 100% / Actual Size. The grid edge is one inch; compare a real small block with a printed cell. If needed, change \blockside in common.tex and rebuild. A poor print fit should not become a mathematical obstacle.

For each K–1 child: 6 blues, 12 greens, 3 purples, and a few reds/yellows. For each middle child: 10 blues, 5 greens, 3 purples. For the upper child: 8 blues, a pencil, and optional string for ribbons. Reuse pieces between boards. For the current 3/3/1 roster, gather **56 blues, 51 greens, and 18 purples**. The optional cutout sheet supplies 12 paper blues. No pink pieces are needed.

Whole-group warmup before separate starting tasks

After a little free handling, bring everyone together around middle Problem 1. Invite children to place blues on Board B. Show how a blue covers two whole triangle spaces, how to turn it, and how to keep pieces inside without overlaps. Then try Board A and notice the remaining gaps. Demonstrate the task and let children contribute pieces; leave the explanation open. An unfinished attempt is not yet evidence that every attempt will fail.

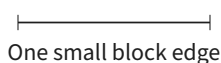
Once everyone has seen the first action, distribute a suitable starting page. This responds to the organizer's report that separate starts at two tables took too much simultaneous management. The revised warmup and packets are proposals to try, not yet tested changes.

Readiness and a flexible hour

K–1: match boundaries and turn pieces; no reading, writing, or arithmetic prerequisite. **2–3:** count spaces through 25 with help; compare up/down orientations and best packings. **4–5:** record arrangements, follow a move rule, and reason about paths and odd/even numbers; small sums help with ribbons. Read any instruction aloud as needed.

0–10: handle materials and explore. **10–15:** common demonstration. **15–35:** investigate. **35–40:** movement/reset. **40–55:** continue. **55–60:** share and tidy.

Give one page at a time. Keep the supporting parent mainly with K–1; the organizer circulates and returns for real discussion with the upper child. Offer hints after attempts. A child can spend the hour on a few useful examples; these pages are not a completion checklist. Keep solutions out of view.

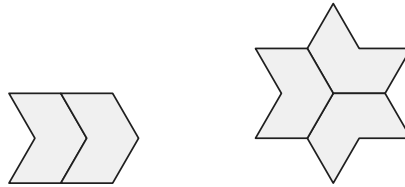


Adult: print **100% / Actual Size**, single-sided. Default edge: 1 inch (25.4 mm). Compare with a 1× green triangle before using these mats.

Blues and empty spaces

K-1 / F01-K-v4

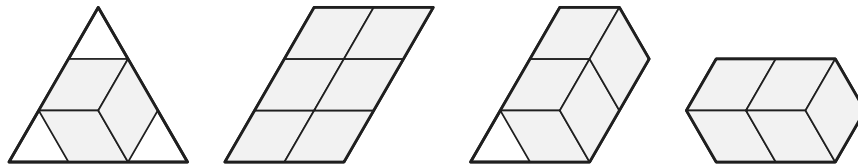
Example fillings: sailboat = 11 greens; cat = 7. Hexagon = 1 yellow, 3 blues, or 6 greens; diamond = 4 blues or 8 greens. Arrow = 2 purples; star = 3. Long hexagon = 5 blues; mountain = 9 greens. Trace pieces before lifting to reuse them.



Ask which piece fits a corner. A checked filling is a useful stop; offer the middle packet for harder work.

Middle Problem 1 / F01-M-v3

Board	Spaces	Up	Down	Most blues	Fewest gaps
A	9	6	3	3	3
B	12	6	6	6	0
C	11	6	5	5	1
D	10	5	5	5	0



Every blue covers two spaces, so an odd total rules out a full filling. C's parity-forced gap matches its 6/5 imbalance. A has the stronger obstruction: each blue uses one up and one down, so only three blues fit. B is the old parallelogram extended down by two rows. B and D both tile.

Keep these explanations out of the launch. If needed later, cover a blue with two greens and turn it: what kinds of spaces does it always use?

Middle Problem 2: side-2 through side-5 triangles

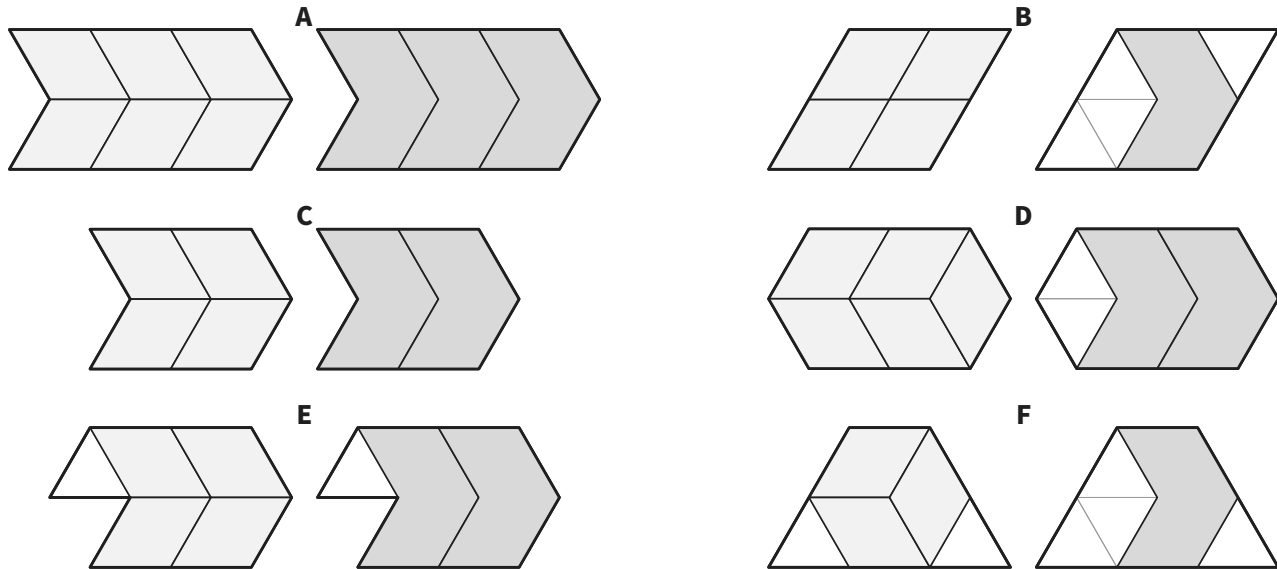
Side length	2	3	4	5
Up spaces	3	6	10	15
Down spaces	1	3	6	10
Most blues	1	3	6	10
Fewest gaps	2	3	4	5

T2 and T4 have even area but cannot tile: even area is not enough. Each horizontal row has one extra up space; a side- n triangle has at least n gaps. Pair each down space with the up space just below-left of it. The remaining ups run along the right edge, attaining exactly n gaps. No formula is needed for this row argument. Count gaps as whole triangle spaces, not pieces.

Compare blues and purples

Middle Problem 3: six boards, two separate trials

Use only blues first, then clear the board and use only purple chevrons. Do not mix pieces in a trial. Every purple occupies four triangle spaces and splits into two blues. Below, each pair shows a best blue packing and a best purple packing; unfilled triangle spaces are the gaps.



Board	A	B	C	D	E	F
Area	12	8	8	10	9	8
Most blues	6	4	4	5	4	3
Most purples	3	1	2	2	2	1
Fewest blue gaps	0	0	0	0	1	2
Fewest purple gaps	0	4	0	2	1	4

Bounds: A and C have full fillings. D has ten spaces, so purples leave at least two; E has nine, so either piece type leaves at least one. F has only three down spaces: at most three blues or one purple (each purple needs two downs), giving two or four gaps. The pictures attain each bound. On B, no purple can cover either sharp corner triangle while staying inside the board (try the six orientations there). So at least two spaces remain; since purples cover four at a time on an eight-space board, at least four remain. One purple fits. Exact whole-cell packing checks independently verify all six maxima.

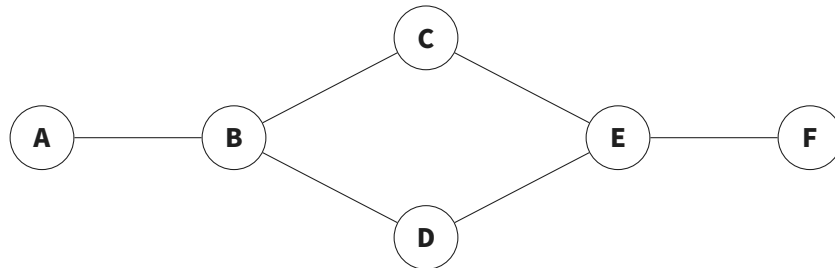
After comparing: ask children to cover one purple with blues. Replacing each purple by two blues preserves the uncovered spaces, so blues can always do at least as well as purples. B and D show that the converse fails; E shows different tile sets can still give the same nonzero optimum. Do not say purple always does worse. The two-pink-triangle task is removed.

Tilings, flips, and routes

Upper Problems 1–4 / F01-U-v3

First give only page 1: one full-size H122 board and eight smaller recording boards. Each blue is recorded by a line between the centers of its two triangle spaces. Keep the bold edge fixed. Let children find and compare tilings before introducing the flip or the six reference cards.

Page 2 shows both fillings of the small hexagon, then highlights the hexagon that changes between A and B. A flip changes exactly those three blues. The six cards are the vertices of the map; children draw connections directly between them. The complete edge list is **AB, BC, BD, CE, DE, EF**.

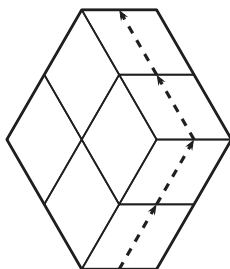


Every card can reach every other. Shortest A–F routes: **A–B–C–E–F** and **A–B–D–E–F**, each four flips. Save “How do you know for sure?” for conversation: A must leave through B, F must be entered through E, and B to E takes at least two moves.

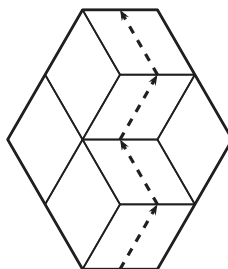
For odd returns, color {A,C,D,F} one color and {B,E} another. Every flip switches color. Three, five, or any odd number of flips cannot return to A. Trying a few routes suggests the result; the two-color argument covers every route. The ribbon gives another proof on the next page.

Upper ribbon problems: tracing before counting

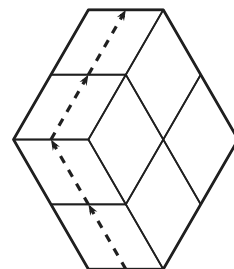
Start at the middle of the bold bottom edge. Cross a rhombus to its opposite horizontal edge, then continue upward. Each step moves left or right. Let children trace with a finger or string before writing the letters. Ask where a second route could branch: each entered rhombus has just one opposite edge.



A: RRLL



B: RLRL



F: LLRR

The total height requires four steps; matching bottom and top centers requires two L's and two R's. Either side of the center is allowed. This is not a stay-on-one-side/Catalan rule. All six orders occur. The cards are a supplied list until the child can explain why the ribbon accounts for every tiling.

What ribbons explain

One tiling, one ribbon; one ribbon, one tiling

Following upward is deterministic. More strongly, trace any rhombus with horizontal edges downward: height strictly decreases until it reaches the sole bottom horizontal edge. Hence all such rhombi lie on this same ribbon. After the ribbon is fixed, every remaining rhombus has the third orientation. Each remaining triangle has at most one partner in that orientation, so the rest of the tiling is forced.

The six orders of two L's and two R's are realized by the six cards. One can list the positions of the R's: 12, 13, 14, 23, 24, 34. Thus there are $\binom{4}{2} = 6$ tilings. Listing the choices is enough; combinations notation is optional.

Card	A	B	C	D	E	F
Ribbon	RRLL	RLRL	LRRL	RLLR	LRLR	LLRR
R-position sum	3	4	5	5	6	7
Possible flips	1	3	2	2	3	1

Exactly the adjacent RL and LR pairs can flip

At a turn, the two consecutive ribbon rhombi and a neighboring third-orientation rhombus fill a unit hexagon. Its other filling changes RL to LR or LR to RL. Straight pairs RR and LL cannot form this hexagon. Conversely every unit-hexagon flip contains the two horizontal-edge orientations, both in the unique ribbon, and changes an adjacent turn. Children can circle the changed pair in the codes and find the same three pieces in the pictures.

A short proof of no odd cycles

Number the four step positions from bottom to top. Add the positions of the two R's: RRLL has sum $1 + 2 = 3$; RLRL has sum $1 + 3 = 4$. One flip moves exactly one R one position, changing the sum by 1 or -1 . Its parity switches every time. A return to the same tiling has the same ribbon and sum, so the number of flips must be even. This proves every closed walk is even without inspecting individual graph cycles.

This score also proves that A to F needs at least four flips, since the sum rises from 3 to 7 and one move changes it by only one. Offer this after the concrete ribbon work, not as a prerequisite for finding a route.

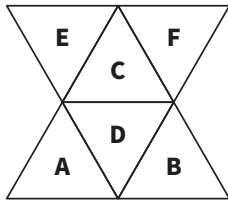
Hints and follow-ups

If a child cannot start, offer only one tiling card to copy. If a flip is hard to see, outline a yellow-hexagon-sized patch. If a map line is uncertain, compare which three pieces change. If a ribbon is confusing, follow the opposite horizontal sides with string. Return to the blocks whenever the letters stop having a visible meaning.

A completed construction, a checked map, or one useful explanation can be a stopping point. Keep optional extension packets for another discussion; the organizer did not use them in the reported meeting. Record the actual cases tried and arguments explained, rather than assuming the whole packet was completed.

Reserves and source notes

Equal up/down counts can still fail / reserve



This six-cell region has three up and three down spaces. A and B can each pair only with D, so a blue tiling is impossible. Blues on A+D and C+E leave B and F empty: two gaps are optimal. This is a matching obstruction, a later question after the simpler area and orientation examples.

Scaling and random moves / reserves

Scale both a small regular hexagon and its three blue pieces by two: its two tilings are unchanged in structure. Keep that larger outline but use $1 \times$ blues: twelve pieces are needed and there are twenty tilings. Changing relative size changes the problem.

Choose uniformly among legal flips on the six-state graph. Long-run visit proportions are $(1, 3, 2, 2, 3, 1)/12$; the mean first positive return to A is 12 moves. Choosing uniformly among four potential centers and staying when the chosen flip is unavailable gives uniform proportions and mean return 6 turns; a stay counts as a turn and an immediate return. These are distinct random rules. The separate extension guide develops them; no probability is needed in the core tasks.

Sources and verification

Teaching: *Math Circle by the Bay*, preface pp. viii–x: substantial themes, clear statements, manipulatives, independent attempts, and flexible pace. Rozhkovskaya, *Math Circles for Elementary School Students*, Lessons 3, 7, 8, “At the lesson”: attempts before representation, checking legal moves, and changing slow recording work after observing children. Lowell *Pattern Block Exercises* – *Google Docs*, p. 1: free play before group questions. Our exact warmup and revised page format come from the organizer’s classroom feedback, not a prescription in these sources.

Geometry and mathematics: Math for Love’s 21st Century booklet, p. 2, identifies the pieces. Danielson’s mosaic and scaling accounts support the nominal one-inch compatibility. Thurston, *Conway’s Tiling Groups* (1990), pp. 757–773, especially p. 767; Saldanha–Tomei, *An overview of domino and lozenge tilings*, §3, Theorems 3.1–3.2; MIT 18.312, lectures 20–23; Levin, *Discrete Mathematics: An Open Introduction*, §2.7. Research and undergraduate notes in `plans/` preserve full references and hypotheses.

The middle-board checker enumerates all grid-aligned blue and purple placements, optimizes disjoint packings, and saves exact witnesses. The lozenge checker enumerates the six upper states and six flips. Student geometry and solution drawings use the same checked coordinates. Source notes, classroom evidence, and draft guidance are in `plans/week-01-classroom-review.md`; `AGENTS.md` changes await organizer review.